Reporting No:01 Week No:01 From : 20/07/26 To : 27/07/26
Project ID: 34
Project Title: SignBridge-AI

WEEKLY REPORT 1

Work done in last week (Attach supporting Documents):

1. Project Planning & Research

- Finalized the project title **"SignBridge AI"**.
- Defined the project scope, objectives, and key features.
- Researched existing sign language translation systems and identified improvements.
- Prepared the complete system architecture and development roadmap.

2. Development Environment Setup

- Installed Python 3.11 and Visual Studio Code.
- Installed Git and created a GitHub repository.
- Created a Python virtual environment.
- Installed required libraries:
  - NumPy
  - Pandas
  - OpenCV
  - MediaPipe
  - Matplotlib
  - Jupyter Notebook
- Created the project folder structure.

3. Python & AI Foundation

- Revised Python fundamentals:
  - Variables
  - Loops
  - Functions
  - OOP
  - File Handling
  - Exception Handling
- Learned NumPy array operations.
- Practiced Pandas DataFrame manipulation and CSV handling.

4. Computer Vision & MediaPipe

- Successfully implemented webcam capture using OpenCV.
- Explored image processing concepts:
  - Image reading
  - Video streaming
  - Frame processing
  - Drawing utilities

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- Integrated MediaPipe Holistic.
- Successfully detected:
  - Hand landmarks
  - Face mesh
  - Pose landmarks
- Studied the structure of landmark coordinates (x, y, z).

5. Model Evaluation and Error Analysis

- Evaluated the trained baseline model using unseen test samples.
- Calculated initial classification performance.
- Generated/analysed:
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - Confusion matrix
- Identified signs that were frequently misclassified.
- Analysed possible reasons for incorrect predictions, including:
  - Similar hand movements
  - Similar hand positions
  - Different signing speeds
  - Insufficient training samples
  - Landmark detection variations
- Documented the initial model performance as a baseline for future improvements.

Reason for incomplete work:

1.
The initial LSTM model was developed as a **baseline model**, and further optimization is required to achieve reliable real-time sign recognition.

2.
Some sign classes showed confusion because visually or temporally similar signs can produce similar landmark sequences. Additional data and model experimentation are required.

3.
The real-time integration of the trained model with the webcam pipeline has not yet been fully completed and will be further developed in the next phase.

Plans for next week:

1. Real-Time Model Integration

- Integrate the trained LSTM model with the OpenCV + MediaPipe webcam pipeline.
- Extract landmarks directly from live video.

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- Convert live landmarks into the required model input format.
- Perform real-time sign prediction.

Target pipeline:

Webcam
↓
OpenCV
↓
MediaPipe
↓
Landmarks
↓
Sequence Buffer
↓
LSTM
↓
Prediction
↓
Sign + Confidence

2. Prediction and Confidence System

- Implement real-time prediction output.
- Display:
  - Predicted sign
  - Confidence score
  - Current FPS
- Establish a prediction threshold to reduce unreliable predictions.
- Experiment with smoothing multiple consecutive predictions.

Example:
Prediction: HELLO
Confidence: 94%

3. Continuous Sign Recognition

- Begin developing the transition from isolated sign recognition to continuous signing.
- Implement a sequence buffer for maintaining consecutive frames.
- Study methods for:
  - Sign segmentation
  - Temporal smoothing
  - Repeated prediction removal
  - Sentence-level sequence construction
- Create the initial pipeline for converting multiple predicted signs into a sentence.

Example:
HELLO
+
HOW
+
ARE
+
YOU

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↓

HELLO HOW ARE YOU

References:

- PyTorch Documentation
- MediaPipe Documentation
- OpenCV Documentation
- NumPy Documentation
- Pandas Documentation
- WLASL Dataset
- INCLUDE – Indian Sign Language Dataset
- How2Sign Dataset
- Research literature on LSTM-based sign-language recognition
- Research literature on sequence classification and deep learning

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